

Live Data Sources & Multilingual OSINT Ingestion Architecture

Target Audience: Management, Directors & Engineering Team Leads

Classification: Confidential — Operational & Technical Briefing

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Executive Summary

***ElectionsSentinel360 (ES360)** is an enterprise, sovereign-grade election operations, situation room command, and legal compliance platform engineered specifically for the Nigerian electoral environment. To provide national visibility across all 36 States and the FCT, ES360 ingests live data streams from multiple field, institutional, media, and digital intelligence channels.*

*This strategic document presents a complete inventory of all operational data sources, followed by an in-depth technical analysis of **Category 3: Multilingual Open-Source Intelligence & Media Streams (OSINT)**. It outlines precise protocols, APIs, webhooks, and automated triage pipelines designed to protect campaign integrity, detect early warnings, and counteract disinformation in 5 Nigerian languages.*

1. Master Inventory of All Operational Data Sources

ES360 aggregates real-time data across six operational categories, bridging rural polling units with the national command headquarters:

Category	Primary Ingestion Channels & Data Types	Operational Impact
1. Polling Units & Field Agents	- Form EC8A Polling Unit result sheets - AI OCR photo pre-fill 176,846 PU agent check-ins & GPS heartbeats - SHA-256 evidence vault (photos, video, audio)	Real-time ground vote tallies & legal tribunal-ready evidence (Section 84 Evidence Act).
2. Official Election Authority	- INEC IReV Portal (<code>inecelectionresults.ng`</code>) feed - INEC Master Geography Database (176k PUs) - BVAS Device Accreditation counts	4-layer PVT result reconciliation & automated §51(2) over-voting mathematical validation.
3. Multilingual Media & OSINT	- Social Media (X, Meta, TikTok, YouTube) - Messaging Hotlines (WhatsApp, Telegram, SMS) - Digital News RSS & GDELT global feed - Live TV/Radio Speech-to-Text (5 languages)	Early warning alerts, sentiment tracking, voter myth detection & rapid counter-disinformation.
4. Incident Command & SOC	- Field emergency distress alerts (violence, BVAS failure) - Automated SLA timers (15m/30m) - SOC security logs & rate-limit alerts	Rapid response dispatch, panic prevention, and system cybersecurity assurance.
5. Compliance & Campaign Ops	108-Position National Campaign Council Roster - Campaign Finance Ledger & spending caps 24-hr pre-election silence period monitor	Electoral Act 2026 legal compliance, ceiling breach alerts & campaign command governance.

6. Post-Election Governance	• Constituent petition & project intake • Electoral promise tracking & sector analytics	Post-mandate delivery tracking across Healthcare, Roads, Education, and Energy.
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2. Deep Dive: Multilingual OSINT & Media Streams

Open-Source Intelligence (OSINT) allows command teams to monitor public sentiment, detect electoral violence threats early, and neutralize viral fake news. Below is the technical breakdown of integration protocols across all OSINT channels.

2.1 Social Media Platforms

Platform	Ingestion Protocol / Method	Technical Details & APIs	Cost & Access Model
X (Twitter)	Official Filtered Stream / Headless Scrapers	WebSocket/HTTP stream querying rules like <i>(#NigeriaDecides OR #INEC)</i> . Unauthenticated backup using `twikit` or `Nitter` RSS bridges.	Enterprise API (Paid) or Free via Academic/NGO Grant.
Facebook & Instagram	Meta Content Library & Graph API	Meta Content Library API for accredited civil society. Graph API `v19.0/{page-id}/feed` for verified news orgs. Meta Ad Library API for political dark ads.	Free upon Meta accreditation / Developer approval.
TikTok	TikTok Research API & RSSHub Scrapers	Extracts public video captions, user profiles, comments, and audio transcript trends under hashtags (<i>#NigeriaElection2027</i>).	Free Research API access for qualified monitoring entities.
YouTube	YouTube Data API v3 & Channel RSS	`search.list` for live election videos (<i>regionCode=NG</i>). Native channel RSS XML feeds (<i>youtube.com/feeds/videos.xml?channel_id=X</i>) for news stations.	100% Free RSS / 10k daily API quota units.

2.2 Messaging & Voter Hotlines

Channel	Ingestion Protocol / Method	Technical Setup & Webhooks	Cost & Delivery
WhatsApp Hotlines	WhatsApp Business Cloud API / Termii API	Dedicated hotline (+234...). Webhook `POST /api/intelligence/webhook/whatsapp` ingests text, voice notes, location pins, and result photos.	Pay-per-conversation (~\$0.015) via Meta or Termii.
Telegram Channels	Telegram Bot API & MTPROTO (<i>Telethon</i>)	Hotline Bot (<i>@ElectionSentinelBot</i>) via Webhook. <i>Telethon</i> background daemon joins public Nigerian political groups to monitor posted claims.	100% Free / Zero API rate limits.
Direct SMS	Termii / Africa's Talking Shortcodes	Provisioned 5-digit shortcodes (e.g. <i>32111</i>). Inbound SMS triggers immediate HTTP POST webhook with sender phone number and carrier data.	Low cost per inbound SMS (~  4/SMS).

2.3 Digital News & Broadcast Streams (5 Languages)

Outlet Type	Ingestion Protocol	Processing Pipeline & Languages	Feasibility
Digital News Portals	Direct RSS/Atom Parsing & Web Scrapers	Node.js `rss-parser` polls top news portals (Punch, Vanguard, Premium Times, BBC Hausa, BBC Yoruba, BBC Igbo) every 2–5 minutes.	100% Free / High Speed.
Global News Database	GDELT Project Real-Time News Feed	Indexes print, web, and online broadcast news in 100+ languages (including Hausa, Yoruba, Igbo, Pidgin). Queryable by geo-location (Nigeria).	100% Free Open Data.
TV & Radio Broadcasts	Live Stream Audio Capture + Speech-to-Text	`ffmpeg` taps live HLS audio streams of Channels TV, Arise, TVC, and FRCN Radio. Slices 30s audio buffers into OpenAI Whisper Large-v3 or Google Speech API for English, Pidgin, Hausa, Yoruba & Igbo transcription.	High Strategic Value / Medium API STT Cost.

3. End-to-End Multilingual Signal Pipeline

All ingested content across social media, hotlines, news feeds, and broadcast transcripts is processed through ES360's backend ingestion pipeline (`apps/api/src/intelligence`):

Step 1: Standardized Data Payload Injection

External collectors POST structured payloads into the NestJS REST API endpoint `POST /api/intelligence/signals`:



Step 2: Automated 5-Language Classification (`language-detector.ts`)

The backend immediately passes the content string to a Unicode-aware heuristic classifier (`\p{L}`) supporting 5 target languages: **English, Nigerian Pidgin, Hausa, Yoruba, and Igbo**. It evaluates marker frequency and assigns a confidence score (0.0 to 1.0).

Step 3: Human Review Routing & Incident SLA Escalation

- **Low Confidence / Mixed Language:** Flagged as needsHumanReview: true and routed to the *Multilingual Intelligence Review Queue* in the Situation Room.
- **High Severity Threats:** Analyst single-click escalates confirmed claims into official **Incidents** with active 15-min or 30-min SLA response countdown timers.

4. Strategic Action Plan & Recommendations for Leadership

1. Immediate API Credential Provisioning (Phase 1):

- Submit applications for **Meta Content Library API** and **TikTok Research API** under institutional observer credentials.
- Register Termii account for Nigerian SMS shortcodes and WhatsApp Business Cloud API webhooks.

2. Deploy Zero-Cost Open Data Collectors (Phase 2):

- Implement the Node.js RSS parser service pulling from 25 top Nigerian news outlets and YouTube channel RSS XML feeds.
- Set up the GDELT open news ingestion pipeline for macro-level state-by-state sentiment tracking.

3. Launch Broadcast Audio Speech-to-Text Pipeline (Phase 3):

- Deploy `ffmpeg` stream recorders for top TV/Radio stations, passing 30-second audio chunks to OpenAI Whisper Large-v3 for automatic Hausa, Yoruba, Igbo, and Pidgin transcript generation.

4. Compliance & Privacy Assurance:

- Ensure all OSINT collectors ingest only public domain data and official hotline submissions.
- Maintain an append-only audit trail (`audit\_events`) for all reviewer actions on intelligence signals.

Report generated by Antigravity AI Assistant for ElectionsSentinel360 (ES360) Command Directorate.